

Introduction to Computer Programming (ICP, CCIT4020)

HKU SPACE Community College, 2021-2022

Assignment 2

(15%)

(Total Marks: 30)

- **Finish this work, based on concepts and techniques learnt in our course.**
- **Students should finish reviewing the related course notes and materials, before doing this assignment.**
- **Individual work: Student MUST FINISH THIS WORK ALONE. Student cannot work with others.**
** Plagiarism / Collusion / Shared work with others are not allowed. Zero mark will be given, with possible disciplinary action.*
- Questions related to program codes are based on Python programming language, unless specified.
- Follow given instructions and guidelines.

Section A, A2A (10 marks)

Multiple Choice (MC) and Matching Questions, Online (SOUL-Quiz Feature)

- Identify and select the option of choice that "best" completes the statement, matches the item, or answers the question.
- **Number of Attempts Allowed: 2**
- **Grading method: Highest Grade**
- **Make sure you have *successfully* completed, "Finished" and submitted your work.**
*** Attempts must be submitted before time expires, or they are NOT counted.**
- Questions related to program codes are based on Python programming language, unless specified.
- 10 MC questions and 10 Matching questions. Each question carries the same mark.

Section B, A2B (20 marks): Programming Tasks

*** Students may reference to the Practicing Labs to complete this task. Refer to the lab documents for details if necessary.**

*** IMPORTANT:**

- Each source code *.py file must ***start with comments***, including the information of individual student (***student name, student id, class number, year of study***) as example below, unless specified.

```
# A0B1.py, ICP Assignment 0
# by CHAN Siu Ming, 20214020, CL02, 2021

print("ICP")
```

- All executable *.py files must be ***terminated with a message displaying*** the information of individual student (***student name, student id, year of study***) as example below, unless specified. E.g. below

```
Hello World!!

.. Done By CHAN Siu Ming, 20214020, CL02, 2021
```

General requirements (unless further specified):

- Students should follow what we have learnt in the course materials, including the lab materials.
- **Unless specified, no data validation required** in the program (program runs in normal situations, and user inputs valid data as expected).
- All related files (including *.py) should be working in the same folder.
- Proper brief **comments** are required, at least at the top of each source code file.
- Proper **indentations** required in writing program codes.

Given Materials:

- This assignment document.
- Python file **A2B1.py**, to be modified and completed by student. Also *modify the top comments for your own Student Information*.

1. A2B1 [10 marks]

Write a Python program for responding **runner condition message** (in a file named **A2B1.py**):

- Ask user (a runner) to answer the following 3 questions:
 - Q1: How long have you been running (km)?
 - Q2: Are you exhausted (y/n)?
 - Q3: Give up (y/n)?
- Based on the user input, show the corresponding response message below.

Q1 (Distance, km?)	Q2 (Exhausted, y/n?)	Q3 (Give up, y/n?)	Response Message
<20	n	y	"Are you sure?"
<20	y	n	"Keep going!"
>=20	n	n	"Great!"
All other Situations			"Doing fine!"

Sample Program Output

```
--- A2B1: Running Condition ---
How long have you been running (km)? 30
Are you exhausted (y/n)? n
Give up (y/n)? n
Great!
```

← User Input

.. Done By CHAN Siu Ming, 20214020, CL02, 2021

```
--- A2B1: Running Condition ---
How long have you been running (km)? 8
Are you exhausted (y/n)? n
Give up (y/n)? n
Doing fine!
```

.. Done By CHAN Siu Ming, 20214020, CL02, 2021

2. A2B2 [10 marks]

Write a Python program for the game of **modified “Rock-Paper-Scissors”** (in a file named **A2B1.py**):

1, 2, 3, 4 replaces



Background description:

Conventional game of “Rock-Paper-Scissors”, with choices Rock, Paper, Scissors

- Rock wins Scissors, Scissors wins Paper, Paper wins Rock. All others are tie/draw.
- Conversely Scissors loses Rock, Paper loses Scissors, Rock loses Paper

Our **modified “Rock-Paper-Scissors”**, with choices **1, 2, 3, 4**

- **1** wins **2**, **2** wins **3**, **3** wins **4**, **4** wins **1**. All others are tie/draw.
- Conversely **2** loses **1**, **3** loses **2**, **4** loses **3**, **1** loses **4**.

Computer's Choice	User's Choice				
		1	2	3	4
	1	Tie	User Loses	Tie	User Wins
	2	User Wins	Tie	User Loses	Tie
	3	Tie	User Wins	Tie	User Loses
	4	User Loses	Tie	User Wins	Tie

Based on the description above, write a Python program for the **game of modified “Rock-Paper-Scissors”** (in a file named **A2B2.py**):

- Ask user to enter the number of plays they want:
 - How many times do you want to play?
- For each play,
 - Ask user the choice from 1/2/3/4 (instead of Rock-Paper-Scissors)
 - What is your choice: 1/2/3/4?
 - Use module **random** and function **random.randint()** to generate a random selection from 1/2/3/4
 - Show the result if it is a win, a loss, or a tie, and record down:
 - * 1 wins 2, 2 wins 3, 3 wins 4, 4 wins 1. It becomes a tie in all other cases.
- After finished all plays, show the overall result as the sample solution.
- **DO NOT set a seed value** and **do NOT call the function random.seed()** of the random generation (even without argument). Each run should be different.

Sample Program Outputs

```
--- A2B2: Modified Rock-Paper-Scissors ---
How many times you want to play? 3
What is your choice: 1/2/3/4? 1
Computer selects 3
... A Tie
What is your choice: 1/2/3/4? 2
Computer selects 3
... You win
What is your choice: 1/2/3/4? 3
Computer selects 4
... You win
Finished All... You win 2 time(s) more than lose.

... By CHAN Siu Ming, 20214020; ICP 4020; 2021 ...
```

```
--- A2B2: Modified Rock-Paper-Scissors ---
How many times you want to play? 3
What is your choice: 1/2/3/4? 1
Computer selects 4
... You lose
What is your choice: 1/2/3/4? 2
Computer selects 2
... A Tie
What is your choice: 1/2/3/4? 3
Computer selects 1
... A Tie
Finished All... You lose 1 time(s) more than win.

... By CHAN Siu Ming, 20214020; ICP 4020; 2021 ...
```

```
--- A2B2: Modified Rock-Paper-Scissors ---
How many times you want to play? 2
What is your choice: 1/2/3/4? 1
Computer selects 3
... A Tie
What is your choice: 1/2/3/4? 2
Computer selects 4
... A Tie
Finished All... An overall Tie.

... By CHAN Siu Ming, 20214020; ICP 4020; 2021 ...
```

3. Evaluation and Submission

- Mark sure students follow instructions, including to follow the required *naming (e.g. file names)*.
- Run, Debug, Test and Evaluate your program based on the requirements.
- Submit program files to SOUL: **A2B1.py**, **A2B2.py**
- **Do NOT compress/zip** or **rename** the files. Submission work not following instructions and requirements may not be marked or be penalized.

*** Remarks:**

- *More testing may be performed during assessment based on the requirements, apart from the given testing cases.*

~ END ~